

ORIGINAL ARTICLE

Detection of unprecedented level of antibiotic resistance and identification of antibiotic resistance factors, including QRDR mutations in *Escherichia coli* isolated from commercial chickens from North India

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Abstract

Aim: This study aimed to investigate the occurrence of antibiotic resistance phenotype and simultaneously understand its genetic basis in *Escherichia coli* isolated from the cloacal swabs of commercial chickens from north India.

Methods and Results: *Escherichia coli* isolates were assessed for susceptibility to 14 different antibiotics using the disc-diffusion technique and were screened for the presence of 22 antibiotic resistance genes (ARGs) by employing PCR. Isolates were found to be highly resistant to fluoroquinolones (nalidixic acid 91%, norfloxacin 73% and ciprofloxacin 66%), tetracycline (71%), beta-lactams (ampicillin 49% and amoxicillin/clavulanic acid 37%), co-trimoxazole (48%), streptomycin (31%) and chloramphenicol (28%); and comparatively less resistant to cefazolin (13%), amikacin (10%), aztreonam (4%), gentamicin (4%) and ceftriaxone (3%). Sixty-three percent of isolates were resistant to more than four different drugs. Abundance of plasmid-borne ARGs like *tetA* (83%), *sul3* (44%), *aadA1* (44%), *strA* (43%), *strB* (41%), *qnrS* (38%), *sul2* (28%) and *aac(6)-Ib-cr* (15%) was observed among the isolates. Forty-five percent of isolates possessed more than five different ARGs. Quinolone resistance-determining region (QRDR) mutations within *gyrA* and *parC* genes were found to be the major determiners of quinolone resistance. QRDR mutations included leu83, asn87 and gly87 within gyrase-A polypeptide and ile80 and lys84 within topoisomerase IV (encoded by *parC*).

Conclusions: Our findings suggest the abuse of antibiotics as feed additives and prophylactic drugs in Indian poultry sector. It also projects this industry as an active hotspot for the replication and selection of ARGs.

Significance and Impact of the Study: Our findings would provide evidence to the authorities for formulating effective strategies for restricting antibiotic usage as non-therapeutic agents in food animals. Occurrence of both plasmid-borne and chromosome-borne resistance towards quinolones can drive movement of resistance phenotype across bacterial species and vertical movement of resistance along the bacterial generations, respectively, which can pose mitigation challenges.

Nitish Rawat and Anjali have equal contributions and both should be considered as first authors.

KEY WORDS

antibiotic resistance, antibiotic resistance genes, chicken, *Escherichia coli*, quinolone resistance-determining region

INTRODUCTION

Antibiotic resistance has emerged as a serious threat to the global healthcare system. It is estimated to cause 700,000 annual deaths worldwide, including 60,000 neonatal mortalities in India alone (O'Neill, 2016). Misuse of antibiotics is the major reason behind the increasing manifestation of antibiotic resistance in clinical and non-clinical settings, and antibiotics consumption in food animals is contributing to the situation (Levy, 2013; Van Boeckel et al., 2015, 2017; Ventola, 2015). Resistance emerging among microflora associated with food animals is still poorly reported from developing countries, leading to incompetent policy formulation to tackle the problem (Aslam et al., 2018; Berendonk et al., 2015; Boeckel et al., 2019).

Consumption of animal proteins is showing growth in most of the developing countries, which has necessitated a rise in antibiotics use as growth promotants and as an alternative to hygiene practices (CDDEP, 2015; Van Boeckel et al., 2015). Poultry provides one-third of the total animal protein worldwide and is also the highest consumed meat type in India (Chatterjee & Rajkumar, 2015; Godfray et al., 2018), making India the fourth major producer of chicken meat (4 million tonnes of meat in 2019) and the fifth major producer of chicken eggs (105 billion eggs in 2019), according to the most recent data (FAOSTAT, 2021). Such a major boom in poultry production is ascribed to the shift from backyard farming to commercial and industrial management of farms in support of heavy investment in antibiotics amended feeds (Chatterjee & Rajkumar, 2015; Van Boeckel et al., 2015).

Antibiotic resistance emerging in the poultry industry has been reported worldwide, including in India (Apata, 2009; Bhushan et al., 2017). Misuse of medically important antibiotics has led to an increase in the incidence of corresponding resistance reported from the poultry sector from low- and middle-income countries (LMICs) (Boeckel et al., 2019). Prevalence of extended-spectrum beta-lactamase (ESBL)-producing and colistin-resistant *Escherichia coli* in live poultry and meat samples have been recently reported from different regions of India (Brower et al., 2017; Ghafur et al., 2019). Another study has shown a significantly higher prevalence of resistance in *E. coli* isolated from UTI patients living near poultry farms in comparison to the patients living away from farms, suggesting towards dissemination of resistance from farms to humans (Sebastian et al., 2021). Such movement of resistance is driven by antibiotic resistance genes (ARGs) which have been reported in microbes isolated from live chickens and meat from various regions of India (Ghafur

et al., 2019; Kar et al., 2015; Olukemi Adesiji et al., 2018; Sivagami et al., 2020). Nevertheless, studies focussed on understanding the abundance of ARGs in farm animals are still lacking. Our study is an attempt towards understanding the prevalence of phenotypic antibiotic resistance and its underlying genetic basis in the microbes isolated from chickens.

We assessed the antibiotic resistance profile of *E. coli* isolates from cloacal swabs of broiler and layer chickens from six farms located in the northern part of India. Focussing on a gram-negative bacterium helped us to provide a picture of resistance to antibiotics that are effective against gram-negative species, as the antibiotic repertoire for gram-negative microbes is limited. *Escherichia coli* itself acts as a sentinel organism for the development of antibiotic resistance as it is a common enteric resident (Brower et al., 2017; Ibrahim et al., 2016; Oakley et al., 2014). Some of its strains are pathogenic and are known to acquire resistance easily (Baker et al., 2018). Hence, a study centring on such an important bacterium will add more value to the inferences drawn afterward.

MATERIALS AND METHODS**Sampling**

A total of six poultry farms (three broilers and three layers) were randomly selected from Sonapat, Haryana (28.9931°N, 77.0151°E), and Rampur, Uttar Pradesh (28.7983°N, 79.0220°E) (India). From each farm, 20 chickens were randomly chosen to take cloacal swabs using aseptic cotton buds. Swabs were immediately transferred to 15 ml conical tubes containing Cary-Blair transport medium (HiMedia). All the samples were transported on ice to the laboratory within 4 h from sampling. Sampling was performed according to the guidelines for poultry/birds facility issued by the Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA), India.

A short interview of each farm owner was also taken to document information on the population of chickens at the time of sampling, their age, weight, expected time and weight at the time they are going to be sold, and the identity of the antibiotics provided in the feed.

Isolation of *Escherichia coli*

As soon as the samples were brought to the laboratory, each swab was streaked on a MacConkey agar plate and incubated at 37°C for 18 h. Streaking was performed in an aseptic

environment (*BSL2* laminar flow hood) in a serially diluted manner. A total of five pink, dough-nut-shaped colonies were picked from each plate and streaked on an M-FC basal plate supplemented with MUG (HiMedia). The plates were incubated at 44.5°C for 24 h. On the following day, plates were observed under long UV. Colonies emitting blue fluorescence were *E. coli* (Eaton et al., 1998). From each plate, four *E. coli* isolates were picked (one each from the previously streaked region) and re-streaked on an M-FC basal plate (supplemented with MUG) and incubated at 44.5°C for 24 h. Three of the four *E. coli* isolates were randomly chosen for antibiotic susceptibility testing after reconfirming their identity using HiIMViC biochemical kit (HiMedia). All confirmed *E. coli* isolates were stored as glycerol stocks at −80°C.

Antibiotic resistance assessment

Kirby-Bauer disc diffusion test

A total of 360 *E. coli* isolates (three per sampled chicken) were selected for susceptibility testing. LB broth medium (5 ml) was inoculated with stored glycerol stock of each isolate (100 µl). It was incubated at 37°C until it reached 0.5 McFarland standard (0.08–0.13 OD at 620 nm). The inoculum was streaked on Mueller-Hinton agar plates using sterile cotton buds. In all, 14 different antibiotic discs were used for susceptibility testing: ampicillin (AMP), amoxicillin/clavulanic acid (amoxycylav) (AMC), cefazolin (CZ), ceftriaxone (CTR), aztreonam (AT), gentamicin (GEN), amikacin (AK), streptomycin (S), tetracycline (TE), chloramphenicol (C), ciprofloxacin (CIP), norfloxacin (NX), nalidixic acid (NA) and co-trimoxazole (COT) (HiMedia). After placing the discs, plates were incubated at 35°C for 15–18 h. Diameter of the zone of inhibition for each antibiotic was measured the next day and each isolate was categorized into resistant, intermediate or susceptible for each drug according to CLSI guidelines for bacteria isolated from animals, 2018, as provided in Table S1 (CLSI, 2018). *Escherichia coli* ATCC 25922 was used as quality control for susceptibility testing.

Extraction of DNA

One isolate out of three that were assessed for phenotypic resistance from each sampled chicken was randomly selected (total 120 isolates) for the screening of ARGs. Glycerol stocks of isolates were inoculated in MacConkey broth overnight at 37°C. The following day, 500 µl of the culture was centrifuged at 3500 *g* (Sigma® 1-14). Pellet was suspended in 500 µl deionized water and boiled for 10 min. The suspension was centrifuged at 1000 *g* (Sigma® 1-14) for 5 min.

Crude DNA in the supernatant was used immediately as the template for the PCR of ARGs.

Screening for antibiotic resistance genes

All 120 *E. coli* isolates were screened for 22 ARGs primarily known to be plasmid-borne (Alekhshun & Levy, 2007). These included major β-lactamase genes (*bla*_{TEM}, *bla*_{CTX-M}, *bla*_{OXA-1}), carbapenem resistance genes (*bla*_{NDM}, *bla*_{IMP}, *bla*_{VIM}, *bla*_{KPC}, *bla*_{OXA-48}), plasmid-mediated quinolone resistance (PMQR) determinants (*qnrA*, *qnrB*, *qnrS*, *qepA*), aminoglycoside resistance genes (*aadA1*, *aac(6)-Ib-cr*, *strA*, *strB*), sulphonamide resistance genes (*sul1*, *sul2*, *sul3*) and TE resistance genes (*tetA*, *tetB*, *tetC*). 2 µl of crude DNA was used in a 25 µl reaction mixture containing 2.5 µl dNTPs (0.25 µM of each dNTP), 2.5 µl KAPA A buffer (1X) (KAPA Biosystems), 1 µl each of forward and reverse primers (10 µM), 15.9 µl of deionized nuclease-free water and 0.1 µl of KAPA Taq DNA polymerase (0.5 U). The name of each primer set, their targeted ARG along with the amplicon size is provided in Table S2.

All oligonucleotide primers were synthesized by GCC Biotech, India. A programmable gradient thermal cycler (PeqStar) from Peqlab (Avantor®) was used to carry out PCR according to the conditions given in Table S2. PCR products were mixed with 3 µl of 6X bromophenol blue and loaded onto 1% agarose gel (w/v) containing ethidium bromide (0.5 µg/ml) and run in 1× TAE buffer at 90V for 50 min. A 100 bp DNA marker (GeneDirex) was used in each gel. The selected PCR products (one representative product for each target gene) were sequenced from a commercially available source (Bioserve) and analysed using BLAST (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>). These products were used as positive controls for the subsequent PCR reactions.

Screening for QRDR mutations

After analysing the phenotypic data of antibiotic resistance, 18 NA-resistant isolates were chosen (three from each farm). DNA was extracted as described earlier and subjected to PCR for *gyrA* and *parC* using primers as described by Yue et al. (2008). Although *gyrB* and *parE* are also known to possess quinolone resistance-determining region (QRDR) mutations, we focussed on *gyrA* and *parC* because of their known prevalence in poultry-associated bacteria (Olukemi Adesiji et al., 2018; Yue et al., 2008). Amplicons were sequenced and analysed using BLAST (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>) and transeq interface (https://www.ebi.ac.uk/Tools/st/emboss_transeq/) to find the possible translated polypeptides. Multiple sequence alignment of the generated polypeptides

was carried out using MEGA6 software (<https://www.megasoftware.net/>).

Data analysis and statistics

Bar graphs and percent fraction graphs were prepared using Microsoft Excel 2013 (Microsoft). On analysis, isolates were found to be resistant to more than one drug and their resistance pattern failed to follow a normal distribution. For making the data comprehensive, a multiple drug resistance (MDR) structure was used with four levels of resistance (0 antibiotics = susceptible, 1 antibiotic = singly-resistant, 2–4 antibiotics = moderately MDR, >4 antibiotics = extremely MDR) (Brower et al., 2017). Comparisons of the resistance profile were made among different farms and tested antibiotics by employing two-way ANOVA using GraphPad Prism 6.01 (GraphPad Software) with consideration of significance value at $p < 0.05$. The relationship between phenotypic resistance and screened genotypes was also studied by employing Pearson's correlation coefficient using GraphPad Prism 6.01 (GraphPad Software) with consideration of significance value at $p < 0.05$.

RESULTS

Broiler chickens were being reared in grow-out houses with space to move around. These farms housed 10,000–13,000 chickens at the time of sampling and were 3- to 5-week old and weighed 0.7–1 kg. These chickens were expected to grow up to 2.5 kg in 10–13 weeks after which they were to be sold off for slaughtering. Floor bedding was poorly maintained with few stretches of husk. Chicken droppings were not cleared regularly, resulting in poor hygienic conditions. Layer chickens were kept in battery cages supplied with feed and water with no space to move. These farms reared 30,000–50,000 chickens which were 13–15 weeks old, weighing 1.3–1.5 kg at the time of sampling. Chicken excreta was being channelled through a poor drainage system into larger tanks, where it was left to settle. Poor maintenance of the channels was responsible for the unsanitary environment. Workers and care-taker were not aware of the basic procedures for maintenance of hygienic conditions in the farm environment. While handling the chickens, workers were directly exposed to the unsanitary environment since they were not wearing gloves or masks. No foot dips/rubs were being taken during entry or exit. Owners agreed to the use of antibiotics in the feeds but were unaware of their identity (except the owner of the farm 2, who agreed to the use of CIP, C and chlortetracycline). Table S3 summarizes the responses provided by the farm owners/managers.

A total of 360 *E. coli* isolates were phenotypically tested against 14 different antibiotics. Highest degree of resistance was observed for NA (91.67%), followed by NX (72.50%), TE (71.11%), CIP (65.65%), AMP (49.11%), COT (47.78%), AMC (36.67%), S (31.11%) and C (28.33%). Below 5% of resistance was observed only against three antibiotics, viz., AT (4.44%), GEN (3.61%) and CTR (3.33%). A collective image of the resistance profile is shown in Figure 1, while farm-wise consolidated data are summarized in Table 1. According to our data, there was no significant difference in the resistance profile between broilers and layers (Figure 1a) (two-way ANOVA, $p \leq 0.05$). Data also failed to segregate based on the geographical locations of the farms (Sonepat vs Rampur) (two-way ANOVA, $p \leq 0.05$).

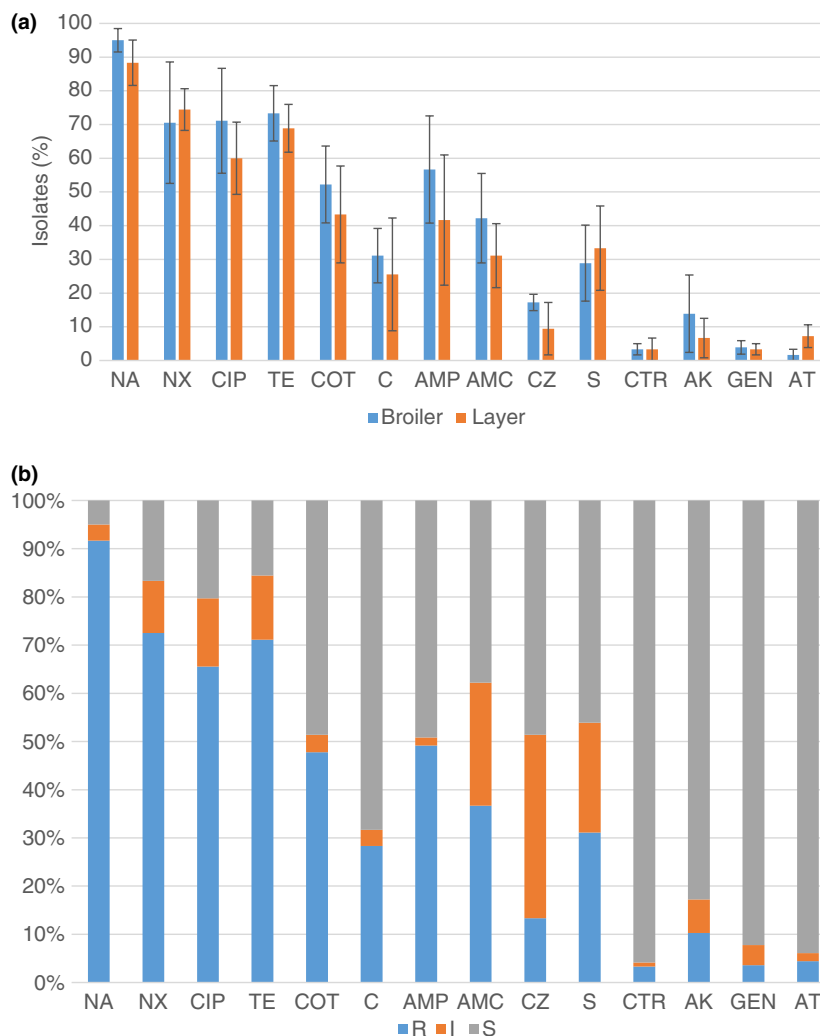
Our data showed a high degree of MDR among isolates from all farms. Collectively, 63.06% of isolates were resistant to more than four drugs (extremely MDR), 31.67% were resistant to 2–4 drugs (moderately MDR) and only 2.22% (8 out of 360) were susceptible to all the drugs tested (Figure 2). Data failed to segregate based on the type of chickens (broiler vs. layer) or their geographical location (Sonepat vs. Rampur) (two-way ANOVA, $p \leq 0.05$). Farm 1 and farm 2 had a very high MDR prevalence where 71.67% and 91.67% isolates, respectively, were extremely MDR (Figure 2).

A total of 120 isolates were screened for 22 ARGs. Out of these, 12 ARGs were reported (Figure 3a). Prevalence of *tetA* among the isolates was the highest (82.50%), followed by *sul3* and *aadA1* simultaneously (44.17%), *strA* (42.50%), *strB* (40.83%), *qnrS* (37.50%), *sul2* (28.33%), *aac(6)-Ib-cr* (15.83%), *bla_{CTX-M}* (6.67%), *tetC* and *qnrB* simultaneously (5.83%), and *tetB* (1.67%). We failed to find any significant difference in the profile of the prevalence of ARGs between isolates obtained from broiler and layer chickens (Figure 3b) (two-way ANOVA, $p \leq 0.05$). Data also did not segregate based on the geographical locations of the farm (two-way ANOVA, $p \leq 0.05$). This indicates at the heterogeneity among farms of the same type and geographical locations.

Multiple ARGs were present in a majority of the screened isolates (Figure 4). The presence of multiple ARGs in isolates varied among farms (Figure S1), independent of the farm type or geographical locations. Data failed to segregate according to farm type or locations and did not follow a normal distribution. Collectively, 97.50% of the total screened isolates across all farms were found to possess more than one ARG (Figure 4). More than 45% of isolates harboured more than four different ARGs and all of the isolates were positive for at least one ARG.

A total of 18 NA-resistant isolates, of which 12 were negative to *qnrS* were screened for mutations within the QRDR of topoisomerase IV (*parC*) and gyrase A (*gyrA*) (Figure S2). In-silico translation of the sequences obtained for *parC* and *gyrA* genes showed a prevalence of QRDR mutations, which have been listed in Table 2.

FIGURE 1 Phenotypic prevalence of antibiotic resistance in *Escherichia coli* isolates. Graph (a) depicts the resistance percentage for all the selected antibiotics of *E. coli* isolates from broiler and layer chickens. No significance was observed in the resistance profile between broiler and layer chickens. Graph (b) shows a collective percent fraction antibiotic resistance profile of all the *E. coli* isolates sampled. Fractions represent the percentage of *E. coli* isolates that show absolute resistance (R), intermediate resistance (I) and no resistance (susceptible – S), which are colour coded accordingly. Antibiotic abbreviations are provided in Table S1



Pearson's correlation coefficient analysis showed a significant correlation between NA resistance and QRDR mutations (Table 3) while no such correlation was found between NA and the presence of *qnrS* gene. Rest of the phenotypic and genotypic pairs that showed significant correlation are shown in Table 3.

DISCUSSION

This study corroborates the claims of the prevalence of antibiotic resistance in food animals, primarily in chicken in India (Brower et al., 2017; Dhanarani et al., 2009; Kar et al., 2015). Farm-raised chickens are known to carry a greater AMR burden, including ESBL genotype than free-range retail chicken (Hussain et al., 2017). A recent study has also reported the presence of colistin-resistant *E. coli* in poultry meat collected from an Indian city (Ghafur et al., 2019). Some reports have found ESBL-producing, multidrug-resistant *E. coli* in poultry, cattle and pigs, possessing various ARGs and virulence markers (Kar et al., 2015; Samanta et al., 2015). Nevertheless, comprehensive studies focussed on resistance and associated

ARGs concerning food animals and their products are still lacking in India. Our study was an attempt towards addressing the paucity of such reports.

Highest level of resistance was reported against fluoroquinolones (NA, NX and CIP), TE, COT (trimethoprim and sulphamethoxazole) and β -lactams (AMP and AMC) which are marked as critically important drugs for human use by the WHO (2017). On comparing the resistance pattern observed in our study to the recently published analysis by Boeckel et al. (2019), we see a higher prevalence of resistance to fluoroquinolones while lower incidence of C, GEN and CTR resistance than the average observed in chicken-associated *E. coli* isolates in Asian LMICs. This can partially be attributed to the cheaper availability of fluoroquinolones than CTR, a third-generation cephalosporin in India.

Owners/managers of all six farms agreed to the use of antibiotics as poultry feed additives. More than 94% of isolates were resistant to at least two different drugs and 63% were resistant to more than four different antibiotics in this study, which suggests the use of multiple antibiotics as feed supplements. Dissemination of multiple-drug-resistant microbes to the surroundings and eventually to the clinical environment

TABLE 1 Resistance profile of *Escherichia coli* isolated from chickens from all six farms. Values in parentheses represent percentage

Antibiotics ^a	Farm 1			Farm 2			Farm 3	
	R	I	S	R	I	S	R	I
NA	60 (100)	0 (0)	0 (0)	58 (96.67)	0 (0.00)	2 (3.33)	53 (88.33)	1 (1.67)
NX	56 (93.33)	3 (5.00)	1 (1.67)	50 (83.33)	7 (11.67)	3 (5.00)	21 (35.00)	15 (25.00)
CIP	52 (86.67)	7 (11.67)	1 (1.67)	52 (86.67)	4 (6.67)	4 (6.67)	24 (40.00)	7 (11.67)
TE	53 (88.33)	4 (6.66)	3 (5.00)	43 (71.67)	8 (13.33)	9 (15.00)	36 (60.00)	8 (13.33)
COT	45 (75.00)	0 (0.00)	15 (25.00)	25 (41.66)	0 (0.00)	35 (58.33)	24 (40.00)	8 (13.33)
C	24 (40.00)	3 (5.00)	33 (55.00)	23 (38.33)	1 (1.67)	36 (60.00)	9 (15.00)	1 (1.67)
AMP	15 (25.00)	0 (0.00)	45 (75.00)	45 (75.00)	0 (0.00)	15 (25.00)	42 (70.00)	2 (3.33)
AMC	15 (25.00)	1 (1.67)	44 (73.33)	41 (68.33)	13 (21.67)	6 (10.00)	20 (33.33)	31 (51.67)
CZ	13 (21.67)	20 (33.33)	27 (45.00)	10 (16.67)	29 (48.33)	21 (35.00)	8 (13.33)	26 (43.33)
S	4 (6.67)	5 (8.33)	51 (85.00)	26 (43.33)	22 (36.67)	12 (20.00)	22 (36.67)	12 (20.00)
CTR	3 (5.00)	2 (3.33)	55 (91.66)	0 (0.00)	0 (0.00)	60 (100.00)	3 (5.00)	0 (0.00)
AK	3 (5.00)	1 (1.67)	56 (93.33)	22 (36.67)	1 (1.67)	37 (61.67)	0 (0.00)	5 (8.33)
GEN	3 (5)	4 (6.67)	53 (88.33)	4 (6.67)	1 (1.67)	55 (91.67)	0 (0.00)	0 (0.00)
AT	0 (0.00)	3 (5.00)	57 (95.00)	0 (0.00)	0 (0.00)	60 (100.00)	3 (5.00)	0 (0.00)

^aAntibiotic abbreviations are provided in Table S1.

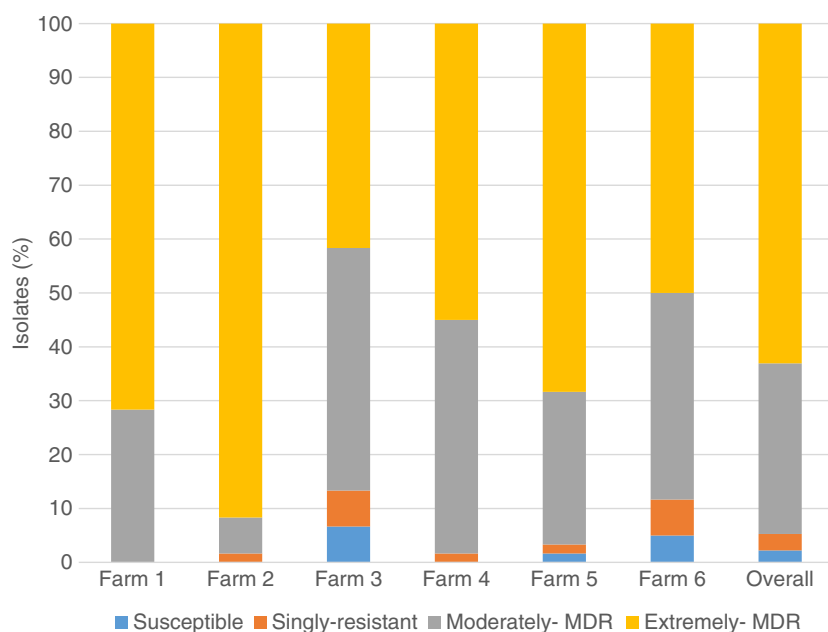


FIGURE 2 Percent fraction graph of *Escherichia coli* isolates for multiple drug resistance according to antibiotic susceptibility testing. Resistance status is defined as (a) extremely multidrug resistant (MDR), if the isolate is resistant to more than four antibiotics; (b) moderately MDR, if the isolate is resistant to 2–4 antibiotics; (c) single, if the isolate is resistant to one antibiotic; (d) susceptible, if the isolate is not resistant to any antibiotic

can put the already limited antibiotic gamut under stress (Chang et al., 2015; Finley et al., 2013). There are numerous cases where resistant strains of bacteria were acquired by humans from food animals either by consumption or by direct contact (Marshall & Levy, 2011). Sometimes, these bacteria can be pathogenic (Fuh et al., 2018). There was no significant interaction of MDR profile among different farms suggesting the lack of a universal regulation of farm management. Such an alarming report of resistance to different antibiotics encompassing different classes and severe MDR makes us suspect that antibiotic administration

in these farms is driven by their unrestricted use (Bhushan et al., 2017).

Genotypic prevalence of antibiotic resistance in animals has been documented in the past (Hu et al., 2017; Kar et al., 2015; Köck et al., 2018; Sharma et al., 2018). Our study has provided a detailed picture of the abundance of ARGs in sampled *E. coli*, suggesting that these farms are hotspots for replication and selection of resistance-conferring genes. *tetA* and *strA* genes, both plasmid-borne, showed correlation with observed TE and S resistance, respectively (Table 3), whereas sulphonamide resistance did not relate well with

Farm 4				Farm 5			Farm 6		
S	R	I	S	R	I	S	R	I	S
6 (10.00)	60 (100.00)	0 (0.00)	0 (0.00)	46 (76.67)	5 (8.33)	9 (15.00)	53 (88.33)	6 (10.00)	1 (1.67)
24 (40.00)	52 (86.67)	3 (5.00)	5 (8.33)	40 (66.66)	3 (5.00)	17 (28.33)	42 (70.00)	8 (13.33)	10 (16.67)
29 (48.33)	48 (80.00)	6 (10.00)	6 (10.00)	26 (43.33)	15 (25.00)	19 (1.67)	34 (56.67)	12 (20.00)	14 (23.33)
16 (26.67)	33 (55.00)	15 (25.00)	12 (20.00)	44 (73.33)	8 (13.33)	8 (13.33)	47 (78.33)	5 (8.33)	8 (13.33)
28 (46.67)	20 (33.33)	3 (5.00)	37 (61.67)	43 (71.67)	0 (0.00)	17 (28.00)	15 (25.00)	2 (3.33)	43 (71.67)
50 (83.33)	35 (58.33)	6 (10.00)	19 (31.67)	2 (3.33)	1 (1.67)	57 (95.00)	9 (15.00)	0 (0.00)	51 (85.00)
16 (26.67)	11 (18.33)	2 (3.33)	47 (78.33)	48 (80.00)	1 (1.67)	11 (18.33)	16 (26.67)	1 (1.67)	43 (71.67)
9 (15.00)	12 (20.00)	24 (40.00)	24 (40.00)	30 (50.00)	14 (23.33)	16 (26.67)	14 (23.33)	9 (15.00)	37 (61.67)
26 (43.33)	1 (1.67)	11 (18.33)	48 (80.00)	15 (25.00)	33 (55.00)	12 (20.00)	1 (1.67)	18 (30.00)	41 (68.33)
26 (43.33)	5 (8.33)	21 (35.00)	34 (56.67)	28 (46.67)	9 (15.00)	23 (38.33)	27 (45.00)	13 (21.67)	20 (33.33)
57 (95.00)	0 (0.00)	0 (0.00)	60 (100.00)	6 (10.00)	0 (0.00)	54 (90.00)	0 (0.00)	1 (1.67)	59 (98.33)
55 (91.67)	0 (0.00)	2 (3.33)	58 (96.67)	11 (18.33)	10 (16.67)	39 (65.00)	1 (1.67)	6 (10.00)	53 (88.33)
60 (100.00)	1 (1.67)	0 (0.00)	59 (98.33)	4 (6.67)	10 (16.67)	46 (76.67)	1 (1.66)	0 (0.00)	59 (98.33)
57 (95.00)	4 (6.67)	1 (1.67)	55 (91.67)	8 (15.33)	0 (0.00)	52 (86.67)	1 (1.67)	2 (3.33)	57 (95.00)

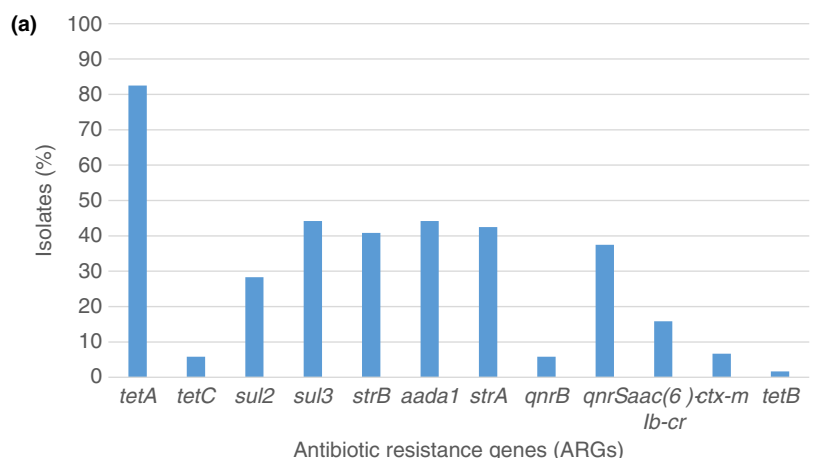
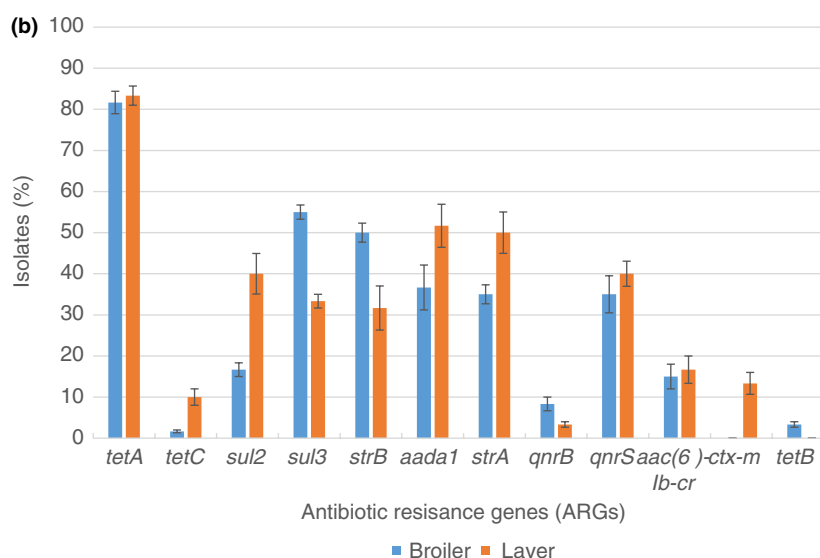


FIGURE 3 Prevalence of antibiotic resistance genes (ARGs) in screened *Escherichia coli* isolates. Graph (a) depicts the percentage of *E. coli* isolates screened positive for selected antibiotic resistance genes. Vertical axis denotes the percentage of positive *E. coli* isolates and horizontal axis denotes different ARGs to which isolates were screened positive. Graph (b) represents a broiler vs. layer profile of *E. coli* isolates which screened positive for selected antibiotic resistance genes. Vertical axis denotes the percentage of positive *E. coli* isolates and horizontal axis denotes different ARGs to which isolates were screened positive (error bars indicate standard error of mean). No significance was observed in ARGs prevalence between broiler and layer chickens



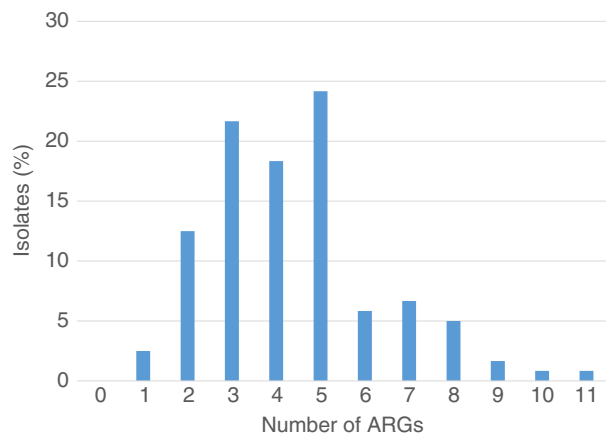


FIGURE 4 Multiple antibiotic resistance genes in *Escherichia coli* isolates. Graph provides a consolidated image of multiple antibiotic resistance genes (ARGs) present in all the screened isolates. Vertical axis denotes the percentage of isolates screened positive for ARGs while horizontal axis denotes the number of ARGs to which isolates were found positive

either *sul2* or *sul3* statistically, suggesting the presence of other sulphonamide resistance genes which we may not have screened, thus requiring further investigation.

Sequencing of *gyrA* and *parC* genes of NA-resistant isolates validates previous reports of QRDR mutations being the primary quinolone resistance determinants (Chen et al., 2021; Zhang et al., 2012) (Table 2). The most common QRDR mutation, which also showed a correlation with phenotypic quinolone resistance reported in our study, was gyrase A-associated leu83 (replacing serine) (Table 3). leu83 has also been reported in MDR *E. coli* isolated from clinical samples, suggesting towards its possible dissemination from the veterinary environment (Pathak et al., 2013). We have also observed PMQR conferred by the *qnrS* gene in 37.50% isolates. Qnr proteins are known to confer low-level quinolone resistance and mediate selection of high-level quinolone resistance among bacteria (Yue et al., 2008). Identification of isolates with similar QRDR mutations and PMQR resistance genes from all six geographically separated farms gives an indication towards the universal abuse of quinolones as feed additives. From an ecological perspective, the occurrence of both types of genotypic features presents an interesting situation where the manifestation of quinolone-resistant phenotype among microbes can be driven horizontally across various species by plasmid-borne resistance and vertically along generations by chromosome-borne resistance. Reports of such microbes from the veterinary sector must serve as an indicator for the possibility of clinical outbreaks, which would pose treatment challenges.

We have also reported *bla*_{CTX-M}-positive isolates from a single layer farm. *bla*_{CTX-M} is known to confer resistance to

cephalosporins, which is in concurrence to the increase in phenotypic cephalosporin resistance among isolates in this farm (25% and 10% resistant to CZ and CTR, respectively). We could not find any positive isolates for carbapenem resistance genes (*bla*_{NDM}, *bla*_{IMP}, *bla*_{VIM}, *bla*_{KPC}, *bla*_{OXA-48}) (Liang et al., 2018), suggesting the unavailability of carbapenems in the feed and prophylactic cocktail of drugs; however, it requires further investigation.

ARGs reported in the present study are more prone to be transferred horizontally due to their known association with plasmids, transposons or integrons (Alcock et al., 2020; Alekshun & Levy, 2007; Carattoli, 2013). Reporting the prevalence of genes encoding ESBL (*bla*_{CTX-M}) and PMQR (*qnrS*) along with strong corresponding phenotypic resistance has reaffirmed earlier claims (Hussain et al., 2017; Olukemi Adesiji et al., 2018). Such studies strive to help institutions to lay out more inclusive and effective policies to tackle AMR (Kakkar et al., 2017; Moudgil et al., 2018). The presence of multiple ARGs among isolates is another serious concern. Interestingly, isolates were found positive for multiple ARGs conferring resistance to the same antimicrobial class (Figure S3), indicating the presence of a complex genotypic heterogeneity to confer similar phenotypic status (Randall, 2004). Hitherto reported ARGs encode proteins showcasing diverse mechanisms to confer resistance. We suspect that bacteria may be employing a combinatorial strategy to achieve robust resistance to antibiotics, which needs to be studied further (Davies & Davies, 2010; Walsh, 2006).

Collection of faecal samples from six farms to finally assess 360 *E. coli* isolates for physical resistance and 120 isolates for identifying genotypic factors of the resulting resistance should provide a good representation of the overall resistance situation in the poultry sector. India has the largest vegetarian population in the world, which translates into a smaller number of organized poultry farms (and other animal farms concerned with meat production) in the country (Appleby & Key, 2016; Chatterjee & Rajkumar, 2015). Most of the organized poultry farms are concentrated within certain regions, which include Sonapat, Haryana and Rampur, Uttar Pradesh. Even within these regions, the number of farms are limited. Lesser number of chickens sampled limits the inferences that can be drawn from this study. These numbers are small due to the reluctance of most of the farm owners/managers from these regions to participate in the study. Owners/managers withholding the identity of antibiotics used as feed additives is another limitation of the study, which requires further investigation.

In conclusion, our study was an investigation to assess antibiotic resistance in poultry. The occurrence of a high degree of resistance against clinically important drugs and the abundance of ARGs among the microbes suggest heavy use of antibiotics as growth promotants. The presence of multidrug-resistant isolates reiterates the seriousness of the

TABLE 2 QRDR mutations within gyrase A (*gyrA*) and topoisomerase IV (*parC*), and PMQR determinants observed in 18 selected nalidixic-acid-resistant isolates

Isolates	<i>parC</i>		<i>gyrA</i>		<i>qnrA</i>	<i>qnrB</i>	<i>qnrS</i>	<i>qepA</i>
	Ser80	Glu84	Ser83	Asp87				
1.1 ^a	Ile	–	Leu	Asn	–	–	–	–
1.2	Ile	–	Leu	Asn	–	–	–	–
1.3	Ile	–	Leu	Gly	–	–	+	–
2.1	Ile	–	Leu	Asn	–	–	–	–
2.2	–	–	–	–	–	–	–	–
2.3	–	–	Leu	–	–	–	+	–
3.1	–	–	Leu	–	–	+	–	–
3.2	Ile	–	Leu	Asn	–	–	–	–
3.3	–	–	–	–	–	–	+	–
4.1	Ile	–	Leu	Asn	–	–	–	–
4.2	Ile	–	Leu	Asn	–	–	–	–
4.3	Ile	–	Leu	Asn	–	–	+	–
5.1	Ile	–	Leu	Asn	–	–	–	–
5.2	Ile	–	Leu	Asn	–	–	–	–
5.3	Ile	Lys	Leu	Gly	–	–	+	–
6.1	–	–	Leu	–	–	–	–	–
6.2	–	–	Leu	–	–	–	–	–
6.3	–	–	–	–	–	–	+	–

Abbreviations: Asn, asparagine; Asp, aspartic acid; Glu, glutamic acid; Gly, glycine; Ile, isoleucine; Leu, leucine; Lys, lysine; Ser, serine; QRDR, quinolone resistance-determining region; PMQR, plasmid-mediated quinolone resistance.

^a1.1 denotes the first isolate of the first farm.

TABLE 3 Phenotypic resistance and genotypic pairs showing significant correlation according to Pearson's correlation coefficient analysis

Phenotypic resistance	Genotype	R^2 value	p value	95% confidence interval
Streptomycin	<i>strA</i>	0.7807	0.0195	0.2544–0.9872
Tetracycline	<i>tetA</i>	0.8346	0.0109	0.3941–0.9906
Nalidixic acid	<i>gyrA</i> QRDR mutation	0.7615	0.0233	0.2091–0.9860

situation. Our results support the immediate need to suspend the use of antibiotics as feed additives and a complete ban on the non-therapeutic use of clinically important drugs in the veterinary sector. However, our investigation requires to be taken further by including more non-clinical environments and non-cultural techniques. Along with the efforts to save the antimicrobial repertoire from wearing off, we must also start searching for the alternatives to antimicrobials.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHORS CONTRIBUTION

Nitish Rawat: Conceptualization (lead), methodology (lead), investigation (lead), formal analysis (lead), writing – original draft (lead) and writing – review and editing (lead). Anjali: Conceptualization (lead), methodology (lead), investigation (lead), writing – original draft (lead) and writing – review and editing (lead). Rohit Jamwal: Investigation (equal), formal analysis (equal) and writing – review and editing (equal). Pukhrambam Pushpa Devi: Investigation (equal) and writing – review and editing (equal). Karuna Yadav: Investigation

(equal) and writing – review and editing (equal). Narendra Kumar: Investigation (equal) and writing – review and editing (equal). Raman Rajagopal: Conceptualization (lead), methodology (lead), funding acquisition (lead), supervision (lead), formal analysis (equal), writing – original draft (lead) and writing – review and editing (lead).

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SUPPORTING INFORMATION

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